

Project Proposal

Document History

Project	The Hand Ram Coffee Press
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Summary of Revisions

Revision	Description of Changes	Revision Author	Date Approved
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1. Description

Coffee machines can take up a lot of valuable counter space in the kitchen, especially in small homes.

The Hand Ram Coffee Press is a small portable device that allows users to easily make a quick cup of coffee in a package that can be stored in a drawer or cabinet. This design has a built-in metal filter that allows users to not have to worry about carrying separate paper filters, and reduces the number of components used in the design.

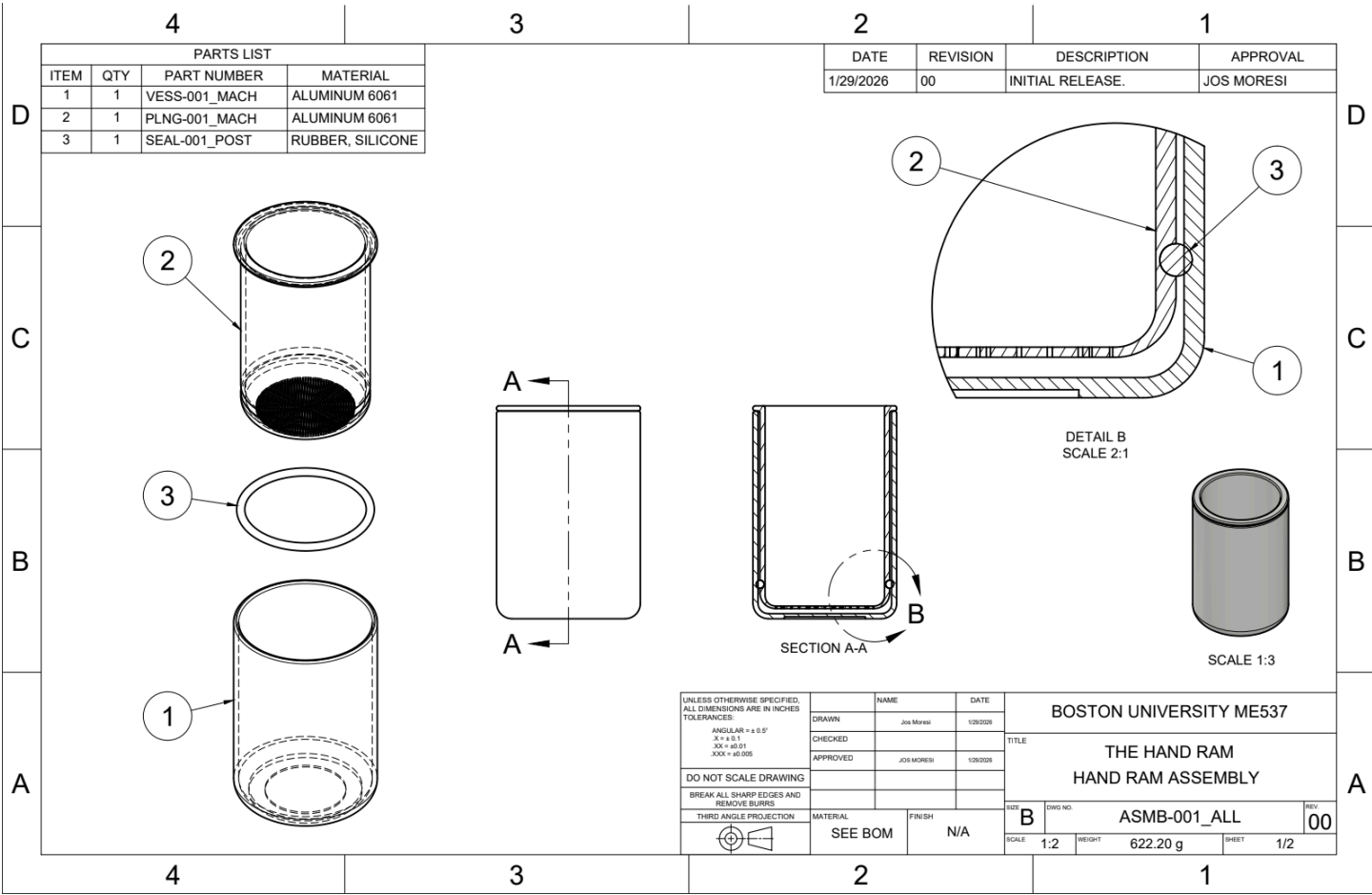
2. Product Requirements

The Hand Ram must meet the following requirements:

1. Must not leak at any point
2. Must withstand liquid temperature up to 100°C
3. Must be food safe
4. Must filter out coffee grinds

3. Parts and Assembly

The Hand Ram will consist of the following components (also found in DOC02):



4. Materials

Aluminum - Used to create both the inner and outer vessels

Rubber - Used for the O-ring seal between the inner and outer vessels.

5. Manufacturing Processes

5.1. Sand casting

Near-net casts for the inner and outer vessels, reducing total raw material per element.

5.2. CNC lathe

Finishing the inner and outer vessels after being cast.

5.3. Water Jet

Cutting small filter holes to separate the liquid from the grinds.

6. Secondary Processes

6.1. Polish

Inside of vessel for proper seal with O-ring

6.2. Stamping/Printing

Labeling for measurements

7. Risks

The following have been identified as potential risks:

- a. Seal may not properly seat in plunger, resulting in coffee leaking through
- b. Aluminum design may conduct a significant amount of heat, resulting in user burning hands
- c. Filter may not be possible to manufacture
- d. Coffee may shoot upwards towards users hand without proper fluid damping through filter
- e. May be discomfort in the lip design for users to drink directly from press
- f. Cost to manufacture first few prototypes may be high

More detail can be found in DOC06 - Risk Management.